

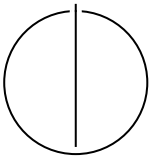
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
FACULTY OF INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification

Lukas Alex Jochim





SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — FACULTY OF INFORMATICS

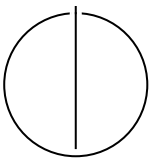
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Decoding Music Perception: EEG-Based Genre Classification

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

Transparency in the use of AI tools

While writing this thesis, Grammarly was utilized to ensure high-quality grammar, style, clarity, and coherence in the content of each chapter. Furthermore, DeepL was used for translation purposes. During the development phase, ChatGPT and Claude Sonnet were used to generate simple code snippets and to gather first ideas for implementing certain functionalities. However, everything generated by one of those tools has been carefully evaluated by me to ensure correctness.

Abstract

Contents

Acknowledgments **iv**

Transparency in the use of AI tools **v**

Abstract **vi**

1 Introduction **1**
1.1 Motivation 1
1.2 Research goals 2
1.3 Outline 2

2 Theoretical Background **4**
2.1 Medical foundations of EEG signals 4
2.2 Brain rhythms 5
2.3 Recording and measurement of EEG signals 6
2.4 Denoising and preprocessing of EEG signals 8
2.4.1 Types of artifacts in EEG signals 8
2.4.2 Artifact Removal techniques 9
2.5 Feature extraction and signal decomposition of EEG signals 11

3 Related work **12**
3.1 Music Genre classification with audio features 12
3.2 EEG datasets and studies with music stimulus 13
3.2.1 DEAP 13
3.2.2 Music BCI 14
3.2.3 MUSIN-G 15
3.3 Emotion classification with EEG signals 16
3.4 Music Genre classification with EEG signals 16

4 Music Datasets **17**
4.1 Audio Set 17
4.2 Million Song Dataset 18
4.3 MagnaTagATune 18

4.4 FMA Dataset 19
4.5 GTZAN Genre classification 20
4.6 Final decision 21

5 Methodology **23**
5.1 Acquisition 23
5.1.1 General setup, environment and stimuli 23
5.1.2 Acquisition device 24
5.1.3 Technical setup 24
5.2 Denoising and Preprocessing 26
5.3 Feature extraction 28
5.4 Classification 28
5.4.1 Support Vector Machine 28
5.4.2 Random Forest 28
5.4.3 XGBoost 28

6 Results **29**

7 Conclusion **30**
7.1 Key findings and results 30
7.2 Problems and Limitations 30
7.3 Future work 30

Abbreviations **31**

List of Figures **33**

List of Tables **34**

Bibliography **35**

1 Introduction

1.1 Motivation

Maybe also mention here the gap between MGR and neurosciences more well, since it will be brought up in the Research goals section The human brain is one of the most complex and important systems within the human body. The brain manages various functionalities like emotion, memory, attention and sensory processing, to name a few. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music is a universal tool to influence one's emotions, concentration and cognition. Furthermore, the brain gives us the unique ability to retrieve many different features out of music, like melody extraction or the identification of an artist or a specific genre. Depending on the style of music, the brain reacts differently. However, although the brain is central to cognitive processes such as listening to music, the core mechanisms remain unknown and are currently black boxes.

To enlighten these black boxes, medical advancements like functional Magnetic resonsance imaging (fMRI) or Electroencephalogram (EEG) arrived, which can measure brain activity. Especially EEG has aroused high interest since they directly measure the neural activity and can be done non-invasive with a little overhead in setup. This is why combining EEG and music has the ability to help better understand how our brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and Computer science.

There is a growing research interest in the affective influence of music on the brain. However, classifying different music genres based on EEG signals remains underrepresented. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science. Especially with the growing investigations on Brain Machine Interfaces (BMIs), adaptive music therapy, personalised music recommendations or neurofeedback systems may become possible. Additionally, entertainment technologies can become more user-centred, facilitating unique experiences that impact many sectors like education and

mental health.

1.2 Research goals

1.3 Outline

To answer the research questions mentioned above, it is vital to have a straightforward research approach, depicted by the structure of this thesis. In the following, a short thesis outline is given, focusing on structure and content in the succeeding chapters.

At first, a theoretical background on EEG signals is given in Chapter 2, building the basis of understanding for all further chapters. After covering the biological procedures behind the generation of EEG signals in the brain, it is described how one can infer the state of the brain based on EEG. In that case, the most prominent brain rhythms are emphasized, describing their impact on the brain and their influence on the measured signal. After the medical background behind the properties of EEG signals, the focus moves towards the three key steps when planning an experiment involving EEG with the classification background. Firstly, the theory behind measuring the EEG data is explained. This section explains what components an EEG system consists of, alongside standard practices regarding the measurement. With the acquisition in mind, the following section deals with preprocessing steps in EEG data, especially on how to deal with artifacts and noise in the raw signal. Once the different types of artifacts appearing have been clarified, different approaches regarding artifact handling are outlined and explained. Last but not least, the last section of the theoretical background covers feature extraction methods in the preprocessed data. Mathematical concepts behind certain extraction functions across all three relevant domains (time-domain, frequency-domain, time-frequency-domain) are broadly described alongside explanations on why these features should be considered.

Chapter 3 then examines the current state of the art in research regarding Music Genre Recognition (MGR) with a focus on EEG. Before investigating research on EEG, a short overview of how one can perform MGR on audio features is provided. This is important to cover as the brain conducts a form of MGR based on the audio signal. Then, a general review is given on EEG datasets and studies that use music to generate the signal. When showcasing these studies, a special focus is given on the experiment setup/environment, measurement setups, and preprocessing techniques. Furthermore, current research is presented on MGR with EEG signals. However, since this is still a relatively young topic in research, this section is accompanied by studies that perform Music Emotion Recognition (MER) on the brain signals. No matter the use case, this section concentrates on feature extraction methods used and on what ML approach the

researchers used for their goal.

Following Chapter 3, Chapter 4 discusses different music datasets that can be used for the purposes of this study. At first, the main requirements for a music dataset being used in this study are explained and clarified. Based on these requirements, five different song databases are presented and evaluated against the mentioned requirements. This chapter then concludes with the final decision on which dataset fits the terms best, thus being used in the study.

With the previous four chapters laying the groundwork, Chapter 5 illustrates the whole process of the study from start to finish. At the beginning, the acquisition of data is emphasized. Before talking about the EEG system used for measurements, the general experiment setup and environment are presented alongside the stimuli in each session. Furthermore, the technical setup is shortly explained by highlighting how each component works together. This section is then followed by a discussion of the denoising steps made on the raw signal, the artifact removal strategy, and the general preprocessing pipeline. Having preprocessed data, the following section deals with the features extracted from the signal. Similar to Chapter 2, each domain is covered, showcasing which features were extracted in which domain. Last but not least, the approach regarding classification is shown. After discussing the general validation and hyperparameter tuning approach, this is further specified by each individual ML used for this project.

Chapter 6 then discusses the results also in regard to the goals of this study. This chapter evaluates the performance of each classification model based on predefined metrics.

Last but not least, Chapter 7 concludes the research project as a whole. It not only sums up the thesis's key findings but also mentions problems and limitations, proposing future research on the basis of this study.

2 Theoretical Background

2.1 Medical foundations of EEG signals

Before one can talk about how to EEG signals can be used for classification or further analysis, it is vital to understand how these signals can be measured non-invasively and how they depict the brain's activity.

The Central Nervous system (CNS) of nerve and glia cells can be found between neurons. Nerve cells can recognize and transmit stimuli through other nerve cells to further locations. To understand how this is possible, one must understand the structure and functionality of a single nerve cell. This kind of cell consists of three main components. At first, there is the cell body with a single nucleus. It is responsible for the cell's metabolism and protein synthesis. Secondly, the Axon, a long cylinder, transmits electrical impulses up to a metre through the body. The third component, dendrites, further supports the transportation of electricity. They are either connected to other dendrites or axons of neighboring cells from which they can receive impulses. Vice versa, it is also possible to transmit the impulse to another cell [13].

Related to the transmission of electrical impulses, the so-called Action potential (AP) is very important regarding EEG, since the AP represents information sent by a nerve. In simple words, AP "is a temporary change in the membrane potential that is transmitted along the axon" [13] caused by the exchange of ions. The AP lasts between 5 and 10 ms, resulting in a peak during the positive polarization of the membrane. After repolarizing to a negative potential, the nerve returns to resting potential after being more negative than the base potential. It is important to remember that an AP is only generated if there is a strong enough stimulus. If the stimulus is not powerful enough, only a small local electrical impulse is created, which is not enough to be transmitted to another nerve, thus not resulting in an AP [13].

As already mentioned, the AP is a key player when it comes to EEG because an EEG signal indirectly records current flows during the excitation of the synapses. During the transmission of an electric impulse because of the activation of neurons, an electrical field is created next to a magnetic field, which can be measured by specialized EEG systems over the scalp (cf. Section 2.3). Since the brain and the scalp consist of multiple layers, only a large population of active neurons can be recognized by systems measuring from the scalp. These must still be amplified for visual inspection since the

recorded signal is very small [13].

The measured signal can differ based on the patient's mental or emotional state. Therefore, one has to try to distinguish the signal into components to make analysis easier.

2.2 Brain rhythms

Visual inspection is traditionally the way to do that when analysing EEG signals for brain disorders or other malfunctions. Specific amplitudes and frequencies of these signals change in a healthy human depending on their state. For instance, the EEG signal differs between wakefulness and sleep. To distinguish the different appearances of the signal, the frequency range is taken into account to form five brain rhythms (cf. Figure 2.1), each of which defines a particular state of the brain [13].

Delta waves occupy the range of 0.5-4 Hz and represent the brain's grey matter [9]. They mainly occur in deep sleep and may be present in the waking state. These signals are very easy to confuse with large muscle artifacts from the jaw or neck, since they are very near the skin's surface. On the other hand, the Delta signals coming from the brain are highly attenuated in passing through the skull [13].

Theta waves lie between 4-7.5 Hz and are primarily present in the thalamic origin. They appear when a human's mental state transitions from consciousness towards drowsiness, which is why they dominate during meditation therapy. [13, 9] Combined with that, it also stands for human growth and serotonin hormones [9].

Within the range of 8-13Hz, Alpha signals are present and are shown as a round-shaped signal. Being the most dominant rhythm in the whole brain and standing for the brain's white matter, it indicates relaxed awareness without any attention and concentration [13, 9]. Studies also show that Alpha waves are a waiting pattern by the visual region of the brain as they are reduced or even eliminated by opening the eyes [13].

Signals with frequencies between 14-26 Hz [13] or 13-30 Hz [9] are classified as Beta waves, which stand for the usual waking state of the brain. Therefore, it is associated with active thinking or problem solving [13]. By inducing the production of Cortisol, it is responsible for learning and memory [9].

Everything above 30 Hz (up to 45 Hz) is considered as Gamma waves. They are very rare in occurrence and are primarily used for detecting and confirming brain diseases [13]. However, they also appear during hyper alertness and the integration of sensory inputs, which is why they combine senses and memory [9].

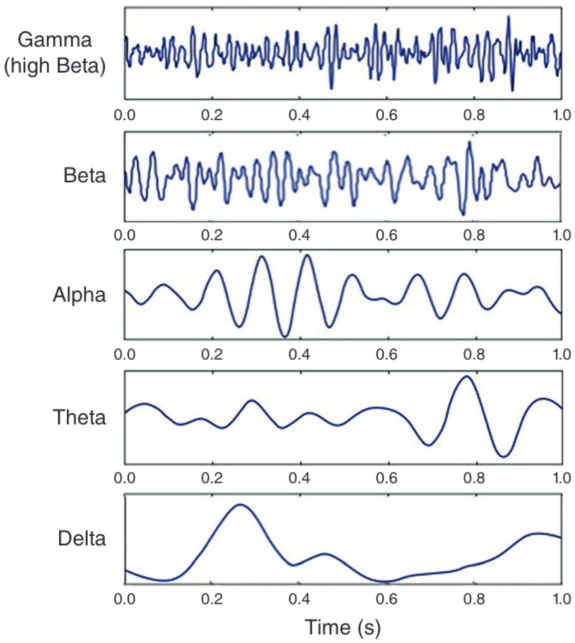


Figure 2.1: The five brain rhythms in EEG [13]

2.3 Recording and measurement of EEG signals

In the previous two sections, it became clear that the electrical activity coming from the APs can be used to depict the neural state of the brain. Therefore, covering how these impulses can be measured is also important.

An EEG system mainly consists of multiple electrodes (called channels), differential amplifiers for each channel, and filters. Multichannel Analogue to Digital converter (ADC) are employed to convert the measured signal points to enable the connection to computerized systems [13].

The most crucial components for data quality are the electrodes. There are different types of electrodes depending on the study's use case (disposable, reusable disc electrodes, headbands, etc.). However, headbands with Ag-AgCl discs are the most popular for multichannel measurements. As already mentioned in Section 2.1, because of the layered anatomy of the scalp, there is high impedance between the cortex and the electrodes, which can lead to deformations in the signal. A rule of thumb is that the impedances should be less than 5 $k\Omega$ and balanced to 1 $k\Omega$ within each other [13].

For unification in measurement and placement in multichannel EEG recordings, the 10-20 system is recommended as a conventional electrode setting. A setup for

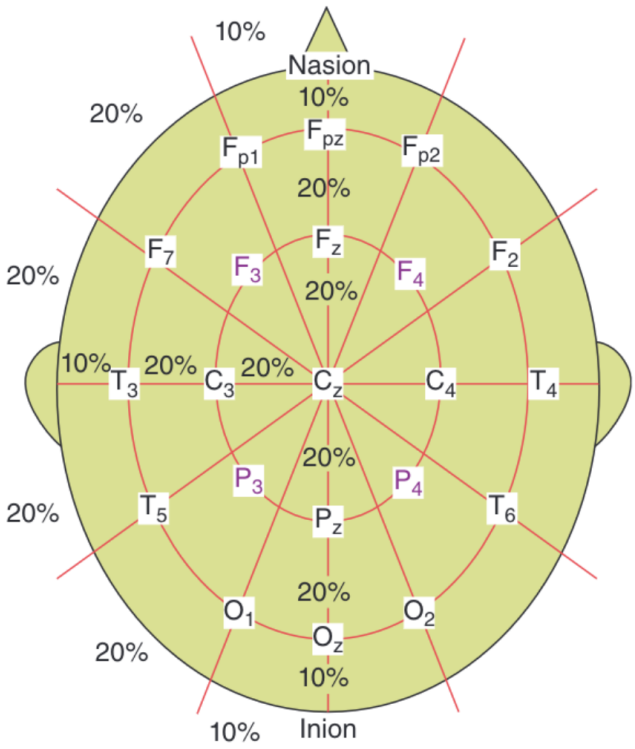


Figure 2.2: Conventional 10-20 EEG electrode positions for 21 electrodes [13]

21 electrodes can be seen in Figure 2.2. This system avoids eyeball placement and employs constant distances between the electrodes. It is called the 10-20 system because the electrodes are placed at intervals of 10 or 20% of the total front-back/left-right distance of the skull. In EEG systems with a larger number of electrodes, the additional electrodes would be placed between the existing electrodes with equidistance between them [13].

As the adequate bandwidth of EEG signals lies around 100 Hz, the minimum sample rate should be 200 Hz to meet the Nyquist criterion[1] Depending on the required resolution, sampling frequencies up to 2000 Hz are possible. In quantification of the signal, 16 bits for each sample is sufficient and popular for many EEG recording systems [13].

[1]The Nyquist criterion states that to accurately reconstruct a signal without aliasing, it must be sampled at a rate at least twice its highest frequency component.

2.4 Denoising and preprocessing of EEG signals

After acquiring the EEG signals, one must follow further steps before performing a classification task. Since EEG is a non-invasive technique, other signals, so-called artifacts, are also measured. Different artifacts are presented before discussing possible methods to remove the data.

2.4.1 Types of artifacts in EEG signals

Physiological artifacts	**Non-Physiological artifacts**
Ocular-related artifacts: eye movements and blinking	Power line interference
Electromyography (EMG) artifacts in head movements	Poor placement of electrode on scalp
Electrocardiography (ECG) artifacts in heart beat	Malfunction of a part in the EEG recording system
Scalp perspiration and movements	Digital artifacts due to loose wiring/-circuit board connections

Table 2.1: Types of artifacts in EEG [7]

Before illustrating what types of artifacts exist, it is essential to mention that artifacts themselves do not destroy EEG signals. They add to the neural activity. Therefore, the types and characteristics must be explained thoroughly before removing any of this data. In Table 2.1, an overview of forms of artifacts can be found. In general, artifacts can be put into two groups: the *Non-Physiological artifacts* and the *Physiological artifacts*. As the name already suggests, while non-physiological artifacts are distortions from the outside world, physiological artifacts are interferences by the human body. [7] These groups will be further described in the following two sections.

Physiological artifacts

As mentioned, Physiological artifacts are characteristic of the human body, including ocular movements, head movements, heartbeats, and muscle noises. For most of these artifacts, it is beneficial to record their corresponding Biosignals. That way, retracing those artifacts in the measured signal is possible. [16]

Ocular artifacts can be measured by Electrooculography (EOG) and are mainly produced by eye movements and blinking. The latter generally produces a much

larger amplitude than the EEG activity. Because of the anatomy of a human, EOG contaminations mostly occur in the frontal electrodes. [16]

An even more common artifact comes in muscular activity, which can be measured in an EMG signal. Appearing in studies where the person is awake, the main actors for these kinds of contaminations are swallowing, talking, and walking, depending on the experimental environment. Most cranial EMG signals are in charge of disadvantageous effects in the recorded signal. However, it is difficult to detect and correct them as EMG not only represents a wide range on the spectral distribution, but also does not repeat, making it even more difficult to characterise. [16]

Last but not least, the activity of the heart (ECG) plays a significant role in contamination of measured EEG signals. Their amplitude is usually low but also depends on electrode placement. In contrast to EMG data, a regular pattern can be visible, which can be further classified by comparing it to ECG measurements. [16]

Non-Physiological artifacts

On the other hand, the so-called non-physiological artifacts are coming from outside-world interference. The most common source is electrical interference from power lines (50 Hz in Europe). This generally can be reduced by shielding the cable or moving the subject far away from the power source. [7]

Another very common non-physiological artifact is the poor placement of electrodes on the scalp as the created DC potential is disturbed. Therefore, electrodes should be checked regularly to ensure high quality recordings. [7]

2.4.2 Artifact Removal techniques

Before presenting state-of-the-art techniques for removing artifacts, it is essential to mention that most of these algorithms and techniques concern physiological artifacts. Controlling the recording environment can correct or altogether avoid most non-physiological artifacts. By checking the placement of the patient and the electrodes on the scalp, non-physiological artifacts remain a small problem in artifact removal compared to handling physiological artifacts. [7, 16]

The first technique is the so-called (Frequency) filtering. It not only removes the power line interference but also removes high- and low-frequency noises. Depending on the goal of the filtering, different filtering techniques are available, mainly a low-pass filter, a high-pass filter, a band-pass filter, and a band-stop filter. While a low-pass filter removes all frequencies above a chosen threshold, a high-pass filter eliminates frequencies below a boundary. The implementation of band-pass filters requires the definition of a lower and upper frequency boundary. When applying, it removes all

frequencies below the lower frequency and above the higher frequency (so keeping all frequencies within the given constraints). Last but not least, a band-stop filter does the complete opposite. It removes all frequencies within the chosen interval, thus keeping the frequencies below the lower and above the upper bounds. Frequency filtering should be the first step of preprocessing EEG signals (even before segmentation) to remove low- and high-frequency noise outside the spectrum of brain rhythms. [7]

When discussing algorithms and techniques dedicated to removing artifacts, assumptions about the recorded signal must be met. The most important assumption is that the recorded signal is the sum of the brain activity and noise. A more generalized version describes the measured EEG signals as linear mixtures of waveforms descending from multiple brain sources and artifacts, ultimately propagating to the scalp. Mathematically, this can be described by the following mathematical equation:

$$\mathbf{X} = \mathbf{AS} + \mathbf{V}, \tag{2.1}$$

where $\mathbf{X} \in \mathbb{R}^{n \times N}$, $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{S} \in \mathbb{R}^{m \times N}$,$\mathbf{V} \in \mathbb{R}^{n \times N}$. While N represents the total count of samples measured, n stands for the number of EEG channels, and m for the number of total sources the signal originates from. In Equation 2.1, $\mathbf{X}$ describes the EEG signals resulting from the measurement. While $\mathbf{V}$ stands for added noise [2], $\mathbf{A}$ is an unknown mixing matrix and $\mathbf{S}$ is a matrix of unknown Sources representing brain sources and artifacts. [16]

With this model, the goal is now to solve the Blind Source Seperation (BSS) problem, so in this case, separating neural activity from activity. For this problem, component-based methods are used because they find principal or independent components equal to the input EEG channels. After processing in the transformed domain, an inverse transformation is applied with the corrected components, resulting in processed EEG data. [16]

In general, BSS aims to estimate $\mathbf{S}$ based on $\mathbf{X}$ in Equation 2.1. It must be mentioned that those methods' effectiveness lies in different assumptions that have to be met to a certain extent, e.g., independence or non-Gaussianity. The more fulfilled these assumptions are, the more effective the BSS method is. Apart from that, the success of such methods also depends on correctly identifying the sources as either brain activity or artifacts, which can serve as a challenge depending on the extent of artifacts. [16]

One of those methods is Principal Component Analysis (PCA), which uses an orthogonal transformation to turn the observed data ot linearly uncorrelated variables (Principal components). The resulting components have the property of having the most significant possible variances while being orthogonal to each other. However, current research suggests that artifacts and cerebral activity can be modeled better

[2] In the following, it is assumed that $\mathbf{V} = 0$

as independent instead of orthogonal. To use this method, Independent Component Analysis (ICA) has emerged as the leading method for BSS. [16]

ICA consists of multiple related methods, separating linearly mixed signals by taking advantage of the statistical independence of the sources. The algorithm usually works based on Higher-order statistics (HOS). After prewhitening the covariance matrix of the data, ICA tries to find a linear transformation $\mathbf{W}$ for estimated sources $\hat{\mathbf{S}} = \mathbf{WX}$. $\hat{\mathbf{S}}$ is then supposed to be as independent as possible. How effective the ICA depends on the independence of sources and the full column rank of the mixing matrix $\mathbf{A}$. Since this algorithm relies on statistical features, the number of samples must be sufficiently large. In most cases, it is recommended to use all available data because cerebral activity and artifacts are spatially stationary through time. [16]

2.5 Feature extraction and signal decomposition of EEG signals

3 Related work

3.1 Music Genre classification with audio features

Before talking about how one can derive music genres from EEG signals, it is important to briefly mention research concerning MGR based on the song itself. This serves as a proof of concept that audio features differ between different genres. To do that, recent research is presented in this section with focus on features and ML models used.

In 2018, Bahuleyan compared the performance of Deep Learning (DL) models with different ML models in labeling audio signals with their predefined genres. While the DL models were solely trained with the spectrogram of each song, the ML models are based on manual feature extraction in time- and frequency domain. The songs come from the Audio Set[1], consisting of seven different genres, including Pop, Rock, Hip Hop, Techno, Blues, Vocal and Reggae. The performance was measured across accuracy, F-score and AUC. Hereby, the Convolutional Neural Network (CNN) has performed best with an accuracy of 64%. Under the ML models, Extreme Gradient Boosting (XGB) performed best amongst Logistic Regression, Random Forest and Support Vector Machine (SVM) with an accuracy of 59%. Nonetheless, this shows that standalone DL models seem to perform better than standalone ML models. However, further evaluation has shown that an ensemble classifier consisting of the best performing CNN and XGB performs best overall with slightly better evaluation scores than the fine-tuned CNN [1].

Similar to the previous study, Haggblade et al. investigated different ML algorithms to classify songs between four different features, being classical, jazz, metal and pop. As a source of songs, they used the GTZAN dataset[2] and filtered all the audio for the four most distinct genres. As features, Mel Frequency Cepstral Coefficients (MFCC) have been used as the only extracted attributes. With a 70 to 30 train-test split, vaious ML models (including SVM, neural networks, k-means and k-NN). It is important to mention that this study employs a Directed acyclic graph (DAG) SVM, where each node represents a two-class SVM trained on each combination of genres. A process of elimination is then used to decide the final classification of the multi-class classifier. When evaluating these models regarding accuracy, the result varied between the

[1] cf. Section 4.1
[2] cf. Section 4.5

algorithms. Although the accuracy also depends on the genre (e.g., most ML models have difficulty classifying Jazz), the researchers found that k-NN and k-means had similar performance of 80% and the DAG SVM reached an accuracy score of 87%. The neural network yieled an accuracy of 96%. According to the study, this large gap between SVM and Neural Networks is surprising as they expected similar performance. [6]

In conclusion, both research papers have proved that the differences in genres are learnable by ML and DL models, showcasing desirable performance in multi-class classification. However, both studies have found out that Neural networks are more superior in performance compared to traditional ML algorithms since they not rely on manual feature extraction [1, 6].

In perspective of the thesis and the goal of classifying music genres based on EEG signals, the fact that ML techniques can identify differences across genres supports the fact that the neural response should be different from genre to genre, which can be captured in the EEG data. These differences and patterns then can be identified by the ML model in theory.

3.2 EEG datasets and studies with music stimulus

As music is a popular stimulus used in EEG studies to collect the signal data, many research projects have been conducted in this area. This section intends to serve as an overview of work that has been done in the context of constructing an EEG dataset where music served as a stimulus. For each dataset, the experiment setup and acquisition methods are described, and certain preprocessing and denoising steps are emphasised.

3.2.1 DEAP

The Database for Emotion Analysis using Physiological Signals (DEAP) is an EEG dataset made public in 2012 by Koelstra et al. It is a multimodal dataset providing data for analysing human affective states. For 32 participants, EEG and peripheral physiological signals were recorded while watching 40 one-minute sections of music videos. The video was also shown alongside the music, serving as an additional stimulus. After each snippet, test subjects were asked to rate each music video in terms of valence (positively or negatively connoted), arousal, liking, dominance and familiarity. [8]

The music comes from a website called Last.fm and was selected using a sophisticated procedure. 120 music videos were initially selected and labelled with keywords out of a list of emotional keywords provided by the Last.fm website. Based on each emotional

keyword, ten songs labelled most with one of them have been selected in the next step of the procedure. The chosen songs were further filtered according to valence and arousal attributes. After this filtering step, a one-minute highlight section was selected with an effective highlighting algorithm from each music video. Last, a web-based assessment experiment was conducted to decide the 40 final music video clips for the DEAP study. [8]

The experiment was conducted in a laboratory environment with controlled illumination. The EEG signals have been recorded with a sampling rate of 512Hz by 32 active AgCl electrodes placed in the international 10-20 system. Apart from that, not only have physiological signals been measured, but frontal face videos for 22 of the 32 participants have also been recorded. The basic procedure of each session includes a five-second baseline recording before displaying the one-minute music video. When the video ended, the participant assessed the clip based on arousal, valence, liking, familiarity and dominance. [8]

For further investigation of the recorded signal data, the researchers provide a preprocessed variant of the EEG data. First and foremost, the data were commonly referenced and then downsampled to 256Hz. The downsampled signal went through a high-pass filter with a 2 Hz cutoff frequency employing the EEGlab toolbox. Eye artefacts were removed by using a BSS technique. For further analysis, the baseline recording of five seconds was removed. [8]

3.2.2 Music BCI

Behavioural responses like in the DEAP dataset can be researched, as well as the perception of musical patterns. Therefore, researchers from Berlin and Copenhagen have chosen polyphonic music as a stimulus for an EEG dataset. Polyphonic music describes music with two or more parallel melodic lines, including different instruments. In their work, the main goal is to explore the suitability of polyphonic music in Brain-Computer Interfacing (BCI). [14]

For this purpose, eleven participants aged 21-50 years were chosen, each not educated in the field of BCI research. For acquisition, a 1000 Hz sampling rate was determined with 64 electrodes placed according to the international 10-10 system. The stimuli were 40-second music clips, each of which contained six tracks. Two of these tracks are meant for stereo audio playback, the other four are a mono mix of the stereo clip recorded as additional EEG channels. Each clip contains three instruments playing musical patterns, where one instrument plays a deviant version while the other two stay the same. Per instrument, there are 3-7 deviations per clip. Synth-pop, which has around 130 bpm, and jazz, which has around 120 bpm, were chosen as genres. While the Synth-Pop clips consist of drums, a bass, and a keyboard, the clips coming from

Jazz comprise flute, piano, and a double bass. Ten different clips were made for both music types with those kinds of deviant patterns. [14]

During the experiment, the participant was seated comfortably in a chair and instructed to stay still and relaxed, especially to minimise eye movements during the trial. The main experiment is divided into ten blocks with 21 music clips played randomly. The listener was told to pay attention to each of the three instruments in seven different music clips. To further ensure attention, the participant has the task of counting the number of deviants in the cued instrument. Before the following clip, the participant must enter the recognised number of deviations. [14]

The signal data were preprocessed to further analyse the recorded EEG data. After downsampling them to 250 Hz, they were low-pass filtered using a Chebyshev filter accompanied by a passband of 42 Hz and a stopband of 49 Hz. The data is sectioned into epochs from -200 ms to 1200 ms for additional denoising. While the prestimulus interval was used for baseline correction, a min-max criterion was employed to discard physiological artefacts. [14]

3.2.3 MUSIN-G

In 2022, researchers published an open-source EEG dataset examining the correlation between the brain and music genres. In the experiment, 20 participants listened to naturalistic songs (meaning mostly without lyrics) across 12 genres, ranging from Indian folk music to Goth Rock or Western electronic music. Apart from the genre, subjective characteristics like familiarity and liking have been considered [11]

It is essential to mention that all participants have little musical training and are therefore considered non-musicians. The participant's EEG data were recorded by a 128-channel high-density EEG cap while sitting comfortably in a dark environment. The main procedure can be divided into two parts. At the start of the acquisition, a baseline recording was conducted during absolute silence lasting for two minutes. During this process, the subject's eyes were closed. The second part then consists of listening to the songs, where the order is randomised depending on the person hearing the audio. A beep sound instructed the participants to close their eyes when the song started to play. After each song, a ten-second gap was chosen, followed by another beep sound indicating to open their eyes for answering two questions. The participant has to rate the song on a scale of one to five regarding familiarity and liking. After answering the questions, the next song was played with a trailing ten-second gap in between. [11]

Since the data is provided as raw and segmented EEG data, the signals can be preprocessed for further analysis. The authors suggest a workflow that includes the EEGLAB toolbox in MATLAB. First, the signals are high-pass linear FIR filtered at 0.2 Hz. This is followed by the CleanLine method, which removes 50Hz of line noise

(comparable to a notch filter). Before segmenting the data into each song, the filtered data were down-sampled to 250Hz. By employing a spectrum criterion, bad channels were removed to prepare them for the artefact removal. The Multiple Artifact Rejection Algorithm (MARA) was used to remove the artefacts, an ML-based method for artefact detection and removal. Lastly, removed channels are interpolated with the spherical function of EEGLAB before the average reference is applied to the whole dataset. [11]

3.3 Emotion classification with EEG signals

3.4 Music Genre classification with EEG signals

4 Music Datasets

Since this project deals with music genre classification, a suitable set of musical stimuli has to be chosen to gather valuable data. As research in the domain of music is widespread, multiple familiar music datasets exist that are used in different studies. Therefore, the difficulty lies in choosing the right dataset for this thesis. However, the project goal crystallises precise requirements for such a music dataset.

First of all, the songs that will be listened to are supposed to be around 30 seconds long. The reason is that the stimulus could be lost if the song runs too long because the brain can adapt to the genre-specific features, thus reducing the visibility of the impact on the brain. Apart from that, the number of songs should be large, with around 800 to 1000 song clips. A large pool of songs is essential to get as much different and versatile data as possible. This helps in future steps regarding classification. However, the genres used in this pool must also meet specifications related to this requirement. First, the songs must be clearly labelled to avoid confusion in later stages of the study. It is also essential to know how balanced the dataset is in terms of genres listened to. This is supposed to be as balanced as possible so that each genre is represented equally. Additionally, it should be easy to employ this dataset in the study. It should be uncomplicated to load the songs and get access to labels or other important information.

In the following, different popular and public music datasets are presented. They will be analysed according to their possible use in this research and how they meet the requirements specified above. After that, the reasons for the final choice of dataset are outlined.

4.1 Audio Set

In 2017, Gemmeke et al. published a dataset for audio events called Audio Set[1]. This is a large-scale dataset of audio events labelled by humans using a carefully structured hierarchical ontology of 632 audio classes. The source of the audio events is derived from 10-second segments of YouTube videos. These audio events are occurrences in the real world, like human sounds, animal sounds, natural sounds, sounds of things, source-ambiguous sounds, environment sounds and music. Then, each sound

[1] `https://research.google.com/audioset/`

is labelled with one ontology and a maximal hierarchical depth of six. In determining labels, the researchers ensured that the tags were intuitive and clearly distinct. [4]

Regarding its use in this project, the audio set has some drawbacks. First of all, this dataset's focus is not music but general audio events. This is why, in the first step, the dataset would have to be filtered to contain only the music clips. However, the fact that the clips are only 10 seconds long means there is a risk that this is not enough stimulus to arouse the brain and see the distinct patterns. Another problem lies in the format of the data. Since every audio event is a YouTube link, each sound must be downloaded from that link to allow seamless integration with the experiment setup, thus illustrating a large overhead. Since the audio event labels range from many different occasions, only the audio events under the category of Music are important in this case. From this subset, only the songs with one of seven genre labels are required: pop, rock, hip hop, techno, rhythm and blues, vocal, and reggae. In total, there would be 40540 clips available. Unfortunately, the songs are not balanced out in terms of genres. [1]

4.2 Million Song Dataset

The Million Song Dataset[2] is a collection of audio features and metadata for a million music tracks, which is also freely available. This is the largest research dataset in the field as it contains 1,000,000 songs, accumulating around 280GB of data. In addition to the songs, mainly from western commercial music in 2011, general metadata and audio features like pitch, timbre, etc., are provided using the EchoNest API. This API is an online resource supporting many music applications on the web. Since most of the "natively" available metadata comes from the EchoNest API, it is easily extendable to further labels like genres. [2]

Although this dataset may seem like a great fit because of its rich amount of songs, the number of songs is too much, since only 800-1000 songs are used during the acquisition, most of the dataset would have to be filtered out. Furthermore, not all songs are trimmed down to a shorter clip but are provided in full length, which does not meet the requirement for this study. Moreover, the genres would have to be added externally since they are unavailable in the native dataset. This also emphasises that it does not guarantee any balance in genres available throughout the 1,000,000 songs.

4.3 MagnaTagATune

One challenge when creating such datasets is determining a label or tag for each song. In the context of genres, how is it decided which genre a song belongs to, as there

[2] `http://millionsongdataset.com/`

could be overlapping features. Based on this hurdle, Law et al. created a music dataset to evaluate different tagging algorithms called MagnaTagATune[3]. It contains 25,863 music clips, each 29 seconds long. These excerpts were taken from 5223 songs. These music pieces range across genres like Rock, Pop, Jazz, Blues, World, Metal, Punk and many more. Each clip is generally accompanied by a vector of binary annotations of 188 tags using the TagATune game. [5]

TagATune is a game with two players that aims to collect music tags from the players. Each player is presented with a music clip, which they are assigned to label. After that, both players see what the other player has chosen as tags. Based on this information, the players must decide whether they listened to the same music clip. [10]

From the perspective of this study, this music dataset has the advantage that each clip is around 30 seconds long, hitting the desired length for one stimulus. Furthermore, it is beneficial that the songs vary in the genres used, thus making them a good candidate for a music dataset. However, there are some drawbacks, especially lying in the labelling of each clip. Although the labelling process is well-thought-out and transparent, each song can have more than one label. Depending on the players, one clip can be annotated with more than one genre tag. And since there are in total 37 genre-related tags [12], deciding which genre is the most suitable choice for a song is very difficult. Related to that, since there are around 26,000 clips, one would have to remove a majority of songs to let it fit into the scope of this thesis. Combined with the fact that multiple genres can be chosen for each song, achieving a balanced dataset in terms of genre is difficult.

4.4 FMA Dataset

The Free Music Archive (FMA)[4] is an open dataset designed for multiple tasks in the domain of Music Information Retrieval (MIR). It consists of 106,574 tracks distributed over 16,341 artists and 14,854 albums. Although this dataset offers many different audio features and metadata about the songs, it is most tailored to music genres. Each track is arranged in a hierarchical taxonomy of 161 genres. As a labelling technique, the researchers labelled the song according to the opinion of the artist who sings the respective song. Therefore, subjective errors in tagging and labelling could be minimised. In addition to that, all tracks are not only stored in simple .mp3 files, but also are fully licensed. To make the use of the FMA dataset easier, different subsets of the complete dataset are offered depending on the research question and scope. In total, there are four subsets one can choose from. The first one is called *Full*, which

[3] `https://paperswithcode.com/dataset/magnatagatune`
[4] `https://github.com/mdeff/fma`

has every track untrimmed and is therefore unbalanced in genre. The *Large* subset contains the same tracks as the *Full* set, but now, the audio is limited to 30 seconds each, extracted from the middle of the track. To tackle the simpler problem of single-label prediction in genres, the *Medium* subset only contains those songs that have only one top genre, resulting in 25,000 30-second audio clips. Furthermore, the number of genres is reduced from 161 to only 16. However, this subset remains unbalanced. Defferrard et al. have constructed the balanced *Small* subset to solve this challenge. It holds 8000 tracks, each being 30 seconds long. From all the genres, the eight most popular ones were chosen, and based on this choice, 1000 clips were chosen for each genre. The eight genres in this case are Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop and Rock. [3]

The FMA dataset brings many favourable characteristics for this research project. The *Small* Subset of this set is a perfect fit since it is balanced according to genres and does not contain that many songs. However, around 900 songs per genre would need to be removed to have a reasonable number of songs for the scope of this thesis. Another advantage is the recency of the publication. It is frequently used in different areas of MIR and is accompanied by many code examples and setup guides.

But, despite of being balanced in regards of the genres, the genres are not 100% clear resulting in unclear behaviour and characteristics of the songs. For instance, the genres International and Instrumental are not comprehensive, since many genres can either have a cultural impact or don't have any lyrics. Another disadvantage is that Classical music is not included, since this type of music is well-known for impacting the brain.

4.5 GTZAN Genre classification

Last but not least, there is the GTZAN Genre Dataset[5], which is a dataset specifically designed for MGR. There are 1000 songs in total originating from 2001 and 2002. Each track is trimmed to 30 seconds and ranges over 10 different genres. These genres are widespread, including blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae and rock. Furthermore, each genre contains 100 songs, resulting in a balanced song set in terms of genres. All audio was gathered from different sources like CDs, radio, microphone recordings and more to vary in the conditions of the recordings. Apart from the audio files, the dataset also contains audio features for each song. [15]

As this dataset is specialised in studies regarding MGR, it is an excellent candidate for the thesis's purposes. With its 1000 songs, it fits perfectly in the right range of songs. Moreover, since each genre consists of 100 songs, it is perfectly balanced, meeting another requirement for the experiment. Another advantage is that the genres are

[5] `https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification`

Dataset	Count of songs	average duration	Genres	Balanced option
Audio Set [4]	40540	10s	7	no
Million Song Dataset [2]	1000000	219s	undefined	no
MagnaTagATune [10]	25,863	29s	undefined	no
FMA Small [3]	8000	30s	8	yes
GTZAN [15]	1000	30s	10	yes

Table 4.1: Overview of potential Music datasets

clearly described, from which clear attributes can be derived. The only drawback in choosing genres is that some genres overlap in characteristics. For instance, jazz originates from blues, thus sharing a lot of common characteristics. Nonetheless, many of the other genres have apparent differences from one another, like classical and metal.

4.6 Final decision

All datasets presented in the previous sections have their strengths and drawbacks. An overview of the key attributes is shown in Table 4.1[6]. Based on the evaluation of each dataset for the project's purposes, the most suitable song source for this thesis will be discussed and explained.

As mentioned, the audio set does not fit the project requirements well. Since its focus is not on music but audio events, applying the set of clips in this use case is difficult. Moreover, ten seconds for each track is insufficient to arouse specific brain activity for each genre.

On the other end, the Million Song Dataset is not beneficial either. Although it only consists of music tracks, they are not trimmed and are full-length. Another disadvantage is the lack of balance in terms of genres. Apart from that, 90% of all songs would have to be removed since only around 1000 tracks are necessary for this study, which can be a significant overhead (especially when ensuring a balanced set of songs).

From all the datasets, the MagnaTagATune dataset has the most sophisticated model of tagging each clip, which tends to higher quality in labels. Another benefit is that all songs are trimmed to 29 seconds, meeting the duration requirement. However, this dataset is not focused on MGR and consists of 188 music-related tags. Moreover, the tags are binary vectors, so some songs could be labelled with more than one genre,

[6]The attributes only correspond to the parts of the dataset that are interesting in terms of MGR. The term undefined is used when there are no clear genre tags or if they depend on subjective interpretation and individual preprocessing

resulting in more difficult handling and unclear distinctions of songs and the genre they belong to. For this reason, the dataset was not chosen for this research.

As the three previous datasets do not seem to fit the requirements, it has to be decided between the small FMA dataset and the GTZAN Genre classification dataset. Both datasets have very similar characteristics as they are specialised in MGR and therefore are balanced in genres, which is one of the most critical requirements in this study. Additionally, all clips are available as audio files with a constant duration of 30 seconds. The key difference lies in the size and genres present in the dataset. While the FMA set contains eight genres with 1000 songs each, the GTZAN contains ten genres with 100 songs each. Besides, compared to the genres represented in the GTZAN dataset, the genres in the FMA collection are not very descriptive and (for some of them) difficult to distinguish. Therefore, the GTZAN song collection is chosen for the experiment setup and as a source of songs. Its easy file structure and naming convention can be seamlessly integrated into any setup, thus being the most suitable fit for this study.

5 Methodology

5.1 Acquisition

5.1.1 General setup, environment and stimuli

The experiment has been conducted in a laboratory setting. As mentioned in Chapter 4, the GTZAN Music dataset is chosen as a basis for the music stimuli. However, for the sake of the quality of the data, it would not be beneficial to hear all 1000 songs in one session. To avoid quick attention loss during a session and gather as much different data under different circumstances, the whole set of audio files is partitioned into 20 distinct sessions, each lasting around 30 minutes. This means that per session, 50 songs are played[1]. For each of these songs, not only are EEG signals measured, but also other biosignals like Photoplethysmogram (PPG), Inertial Measurement Unit (IMU) and temperature. However, this work will solely focus on the recorded EEG data. However, these 50 songs are not chosen randomly. Five songs from each genre are being played during the session to ensure that each of the ten genres from the dataset is represented in each session. This also ensures no genre bias in the data, thus resulting in a balanced signal across genres. During listening, the order of the songs is randomised to evoke a brain reaction that is as natural as possible, as the brain cannot predict the genre that is being listened to. Additionally, a five-second break is implemented in between songs to not overstimulate the brain.

All 20 sessions have been recorded over two weeks, with an average of two sessions per day. Between each session, there was a short break of between ten and 30 minutes. This verifies a sufficient pause of the brain to fully prepare for the next session. The test person is the author, a 21-year-old, right-handed college student without any common (brain) diseases. To further guarantee a high quality in the acquired signal data, the experiment's environment and setting must be adapted. First and foremost, the test person is instructed to sit comfortably in a chair with his eyes closed to minimise artificats coming from blinking and eye-movement. Furthermore and most importantly, any movements, especially in the head region, must be limited and as minimal as possible since the EEG device reacts very sensitive to head movements. Apart from that, the room was quiet and isolated with dimmed natural lighting combined with the

[1] Because of a corrupt audio file, one session only consists of 49 songs

brightness from a desktop monitor. The songs were played over the EEG headset (cf. Subsection 5.1.2) with average and constant volume.

5.1.2 Acquisition device

A non-traditional approach is chosen to record the EEG signal data coming from the brain. For this study, a prototype for a hybrid biosensing platform *Hugin* by Jabra is chosen. It is a Bluetooth headset that can play music and measure biosignals at the same time. For Audio, it uses classic Bluetooth to connect to a standard PC or Laptop. For the data, however, it uses a second low-energy bluetooth channel for the biosignal data, which can be connected to a second data acquisition PC.

The headphones have multiple sensors and electrodes to measure biosignals like EEG, PPG, IMU and temperature. For EEG, the headphones are supplied with 2 × 4 sensor electrodes and two bias electrodes. The sensors are easily configurable and provide an ultra-low-noise front-end, meaning that the data points are amplified with minimal distortion and interference. Besides that, the electrodes sample with 24-bit and a frequency of 250Hz.

For PPG, one optical sensor on the right side provides a high resolution of 19-bit ADC with a configurable sample rate. They also support raw data output or built-in algorithms to conclude heart rate or SpO2.

The IMU data is sensed by a 3-axis digital accelerometer fused with a 3-axis digital gyroscope. On a full scale, the accelerometer can recognize acceleration from $\pm 2g$ up to $\pm 16g$ (in steps of the power of two). The gyroscope can measure angular velocity from $\pm 125°/s$ up to $\pm 4000°/s$. The recognised data is then combined to provide accurate information on orientation, movements and position.

Last but not least, there are three digital thermometers, one on the skin, another one inside the earcup and one on the outside of the earcup to measure ambient temperature. All of those temperature sensors have an accuracy of ± 0.1°C.

Since this headset is still a prototype, they still is ongoing development and research on the headset. Nonetheless, some small-scale research projects already employed the biosignals from the headphones. These projects range from stress detection using the PPG data to device control and the recognition of cognitive fatigue using the EEG sensors.

5.1.3 Technical setup

Multiple technical components have to come together to gather the experiment's data. All the components employ the Lab Streaming Layer (LSL). It provides a standardised way to stream time-sensitive data between devices and applications. Devices or apps

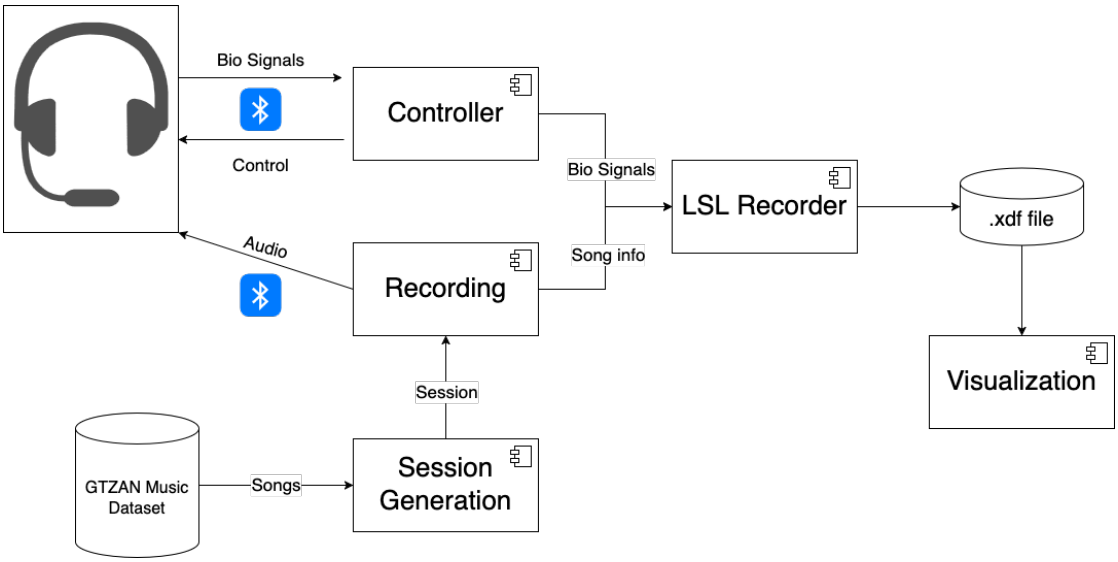


Figure 5.1: Technical setup

then push data on this stream, which is captured and stored in a special file format. The workflow is illustrated in Figure 5.1 and will be explained in the following.

At the core, there are two main programs responsible for the data. On the one hand, there is the Headset platform called *BioGNee*. This program is responsible for the sensor setup and starting and stopping the signal acquisition. Communicating with the headset's low-energy Bluetooth Channel reliably sets up the biosensors of the hybrid headphones, making the device ready for acquisition. After a successful setup, the data is pushed on the corresponding LSL Stream. On the other hand, the study uses a custom Python script. This Python script has two main tasks. Firstly, it loads all the songs from one of the 20 pre-generated sessions and randomly shuffles them. After that, the program plays each song one by one. Simultaneously, the script also pushes timestamps during recording to assign each segment to each song later.

The LSL Recorder combines both of these components. The LSL recorder collects all open streams on which the device/programs push their data. These are collected and stored in an XML-like file called .xdf. The resulting document, therefore, provides a unified way to not only store the signal data but also the respective time markers. These can be visualised by a custom MATLAB script to inspect the data quality. Important to note is that each distinct session occupies one .xdf file.

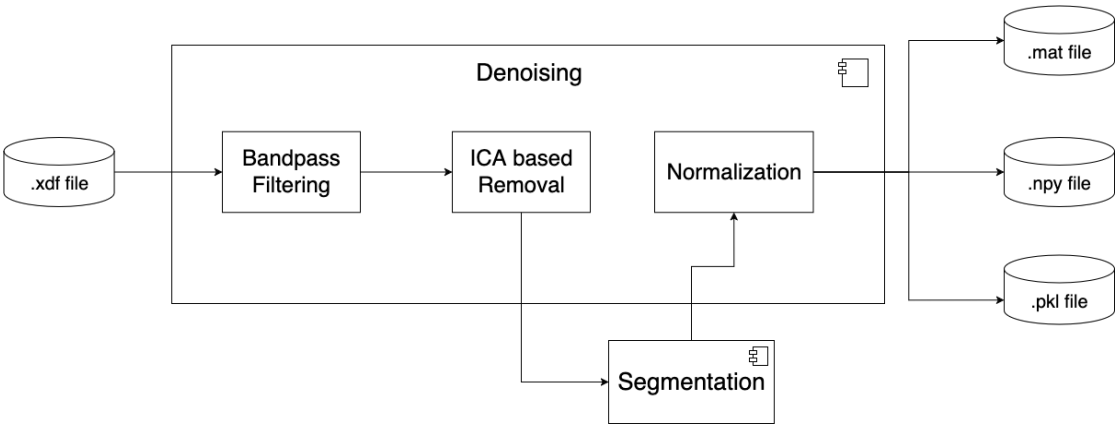


Figure 5.2: Preprocessing of the raw EEG signal

5.2 Denoising and Preprocessing

After measuring and recording the EEG activity, the next step is to preprocess the acquired signal. Apart from segmenting the data into their respective song listened to, the signal has to be denoised, which is the most essential step. As mentioned in Section 2.4, many different artifacts besides the neural activity appear during recording. Therefore, it is crucial to recognize and omit most of these contaminations to have valuable data for feature extraction and classification. The complete preprocessing pipeline is visualized in Figure 5.2, where each step is further explained in the following.

Before explaining how the data has been denoised, it is vital to investigate the state of the raw measured data. An example can be found in Figure 5.3. Most notably, there are significant amplitude shifts over the 30-minute recording sessions. Depending on the channel, there is either an increasing or a decreasing trend in amplitude. This can be reasoned with the fact that the electrodes are DC-coupled in the system. The drift then happens due to temperature changes or skin impedance.

To overcome this signal drift, the first step of the denoising pipeline is to apply a bandpass filter on the whole session. As boundaries, 1 Hz and 30 Hz are chosen as the lower and upper borders, respectively. This handles the mentioned drift and filters out all high and low frequency noise that does not match a certain brain rhythm.

After removing most of the frequency-related noise, the artifacts in the EEG data must be detected and removed. An ICA algorithm is applied to the session data. After prewhitening the data, an ICA is applied, splitting the EEG matrix into a mixing matrix accompanied by the sources. Since eight EEG channels are present, the ICA algorithm was performed with eight components. As a next step, it has to be decided whether a component is an artifact or an actual source of cerebral activity. This project deals

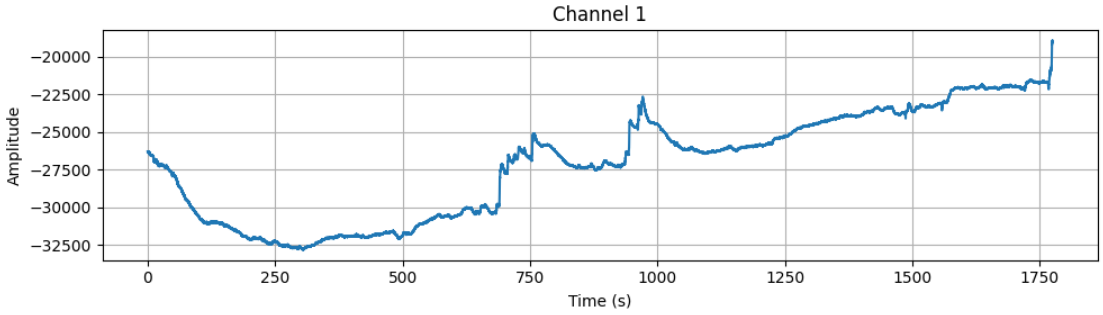


Figure 5.3: Raw signal of Session 1

Rejection Criterion	**Threshold**
Peak-to-Peak Amplitude (PTP)	85th percentile of all components
Kurtosis	85th percentile of all components
Skewness	85th percentile of all components
Autocorrelation (lag 1)	85th percentile of all components
Relative Delta Band Power	$>$ 0.5 of total power
Relative Beta Band Power	$>$ 0.35 of total power
Mixing Matrix Sparsity	$<$ 3 active channels (relative threshold = 0.2)

Table 5.1: ICA Component Rejection Criteria

with an unconventional way of measuring the EEG data, so custom criteria must be determined for this task. In Table 5.1, the rejection criteria for each component can be seen. PTP, skewness, kurtosis, and autocorrelation are used to detect unusual signal characteristics deviating from brain activity. The spectral power thresholds, including the Delta and Beta Power ratio, find artifacts by finding components dominated by specific frequency bands. Lastly, by investigating the sparsity of the mixing matrix, it is evaluated how widely a component affects the EEG channels. Suppose a component is recognized as an artifact. This component is set to zero before converting the mixing matrix with the now cleaned sources to the artifact-removed version of the original EEG data. This results in a signal where most of the high amplitudes are removed with a much more consistent EEG signal, thus suggesting that most of the unwanted artifacts have been removed.

With the cleaned EEG signal at hand, one can now segment the signal into its 50 individual songs[2]. Since the start and the end of one song are pushed to the LSL stream,

[2]The custom Python script also supports the segmentation of biosignals other than EEG. However, these will be ignored in the following as this thesis focuses on EEG signals

the timestamp of those markers is read out to only keep the signal data appearing in this interval. This is done over each channel, thus resulting in a table with each song occupying a row and the channels as columns.

To further improve the data quality for supporting good results in a classification task, each segment is z-normalized with the formula $Z = \frac{X-\mu}{\sigma+\varepsilon}$ so that the data at hand has zero mean and a unit variance.

After all these steps, the denoised, segmented, and normalized data is stored in three different formats, being .mat, .npy, and .pkl, where each session is held in one file across all file formats.

5.3 Feature extraction

5.4 Classification

5.4.1 Support Vector Machine

5.4.2 Random Forest

5.4.3 XGBoost

6 Results

7 Conclusion

7.1 Key findings and results

7.2 Problems and Limitations

7.3 Future work

Abbreviations

fMRI functional Magnetic resonsance imaging

ML Machine Learning

BMI Brain Machine Interface

CNS Central Nervous system

AP Action potential

ADC Analogue to Digital converter

BCI Brain-Computer Interfacing

BSS Blind Source Seperation

MARA Multiple Artifact Rejection Algorithm

EMG Electromyography

ECG Electrocardiography

EOG Electrooculography

PCA Principal Component Analysis

ICA Independent Component Analysis

HOS Higher-order statistics

DL Deep Learning

CNN Convolutional Neural Network

XGB Extreme Gradient Boosting

SVM Support Vector Machine

MFCC Mel Frequency Cepstral Coefficients

DAG Directed acyclic graph

PTP Peak-to-Peak Amplitude

FMA Free Music Archive

MIR Music Information Retrieval

EEG Electroencephalogram

PPG Photoplethysmogram

IMU Inertial Measurement Unit

LSL Lab Streaming Layer

MGR Music Genre Recognition

MER Music Emotion Recognition

DEAP Database for Emotion Analysis using Physiological Signals

List of Figures

2.1 The five brain rhythms in EEG [13] 6
2.2 Conventional 10-20 EEG electrode positions for 21 electrodes [13] 7

5.1 Technical setup 25
5.2 Preprocessing of the raw EEG signal 26
5.3 Raw signal of Session 1 27

List of Tables

2.1 Types of artifacts in EEG [7] 8

4.1 Overview of potential Music datasets 21

5.1 ICA Component Rejection Criteria 27

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